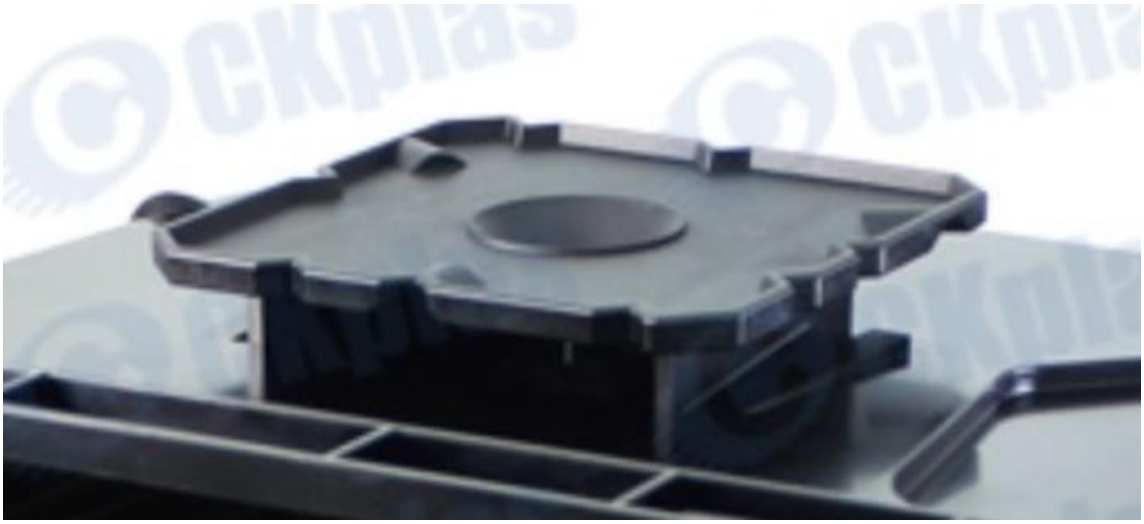


Review on Pose Estimation using Computer Vision of intern project

Tzu-Chieh Tai

Cover image source: CKPlas, *FOUP Frames*, accessed August 26, 2024, www.ckplas.com.

1 Purpose

This work aims to replace the existing localization method that uses Apriltag by computer vision while still providing high accuracy.

2 Related work - Paper review

Many approaches have been devised to estimate 6D (x, y, z, pitch, roll, yaw) pose. He [1] classified these methods into learning-based methods and non-learning-based ones. To choose the best techniques suitable for our mission, I list the tools mentioned in the review below:

2.1 Templates matching

By matching real-world situations with templates that have the highest similarities, the 6D pose can be obtained or more precisely determined by solving PnP. This can be done in 2D or 3D, with the number of templates determining the accuracy of the estimation.

2.2 Key points matching

Neural networks, primarily used in learning-based methods, are employed to locate key points. Transformation relationships are determined by comparing two sets of key points: those from ground truth data and those detected.

2.3 Holistic estimating

Directly applying neural networks to predict 6D pose from 2D or 3D images turns the estimation problems into classification tasks, often leading to poor accuracy.

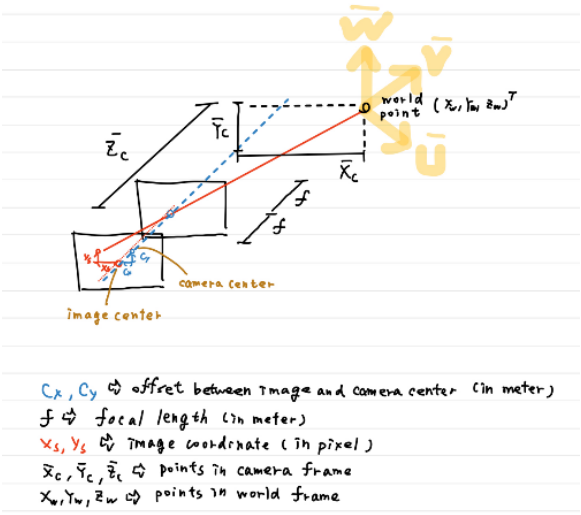
2.4 Features matching

Used in non-learning-based methods, certain mathematical operations, such as SIFT, SURF, and PCOF, are adopted to extract characteristics or point clouds from images. These characteristics can then be further processed through template matching, ICP, or other methods to determine the transformation parameters.

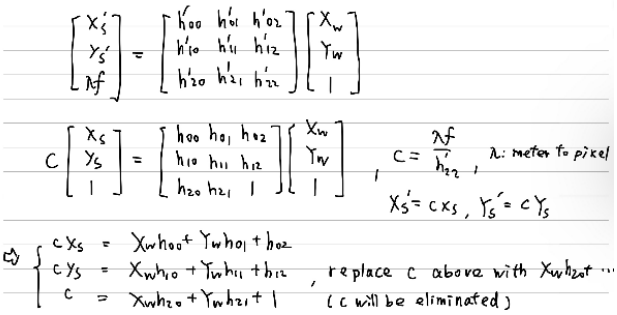
Among all methods, pose estimation using RGB-D information achieves the highest accuracy, making it a strong candidate to replace AprilTag.

3 The Current Solution – Apriltag

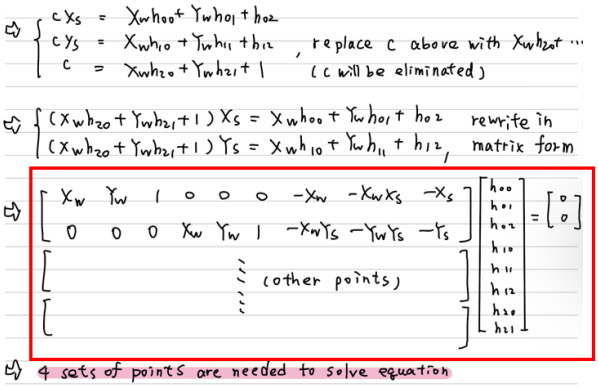
Compared to the heuristic and regressive numerical methods mentioned above, AprilTag's principle—homography—focuses on finding an analytical solution for transformation parameters. The 3x3 homography matrix, similar to the homogeneous transformation matrix (HTM) used in robotics, has one column eliminated for computational convenience. Rotational and translational relationships are determined by solving this matrix using Singular Value Decomposition (SVD) once an image is provided. The workflow of AprilTag's pose calculations is illustrated in the images below.



➢ Construcitng pin hole camera model (created by the author)



➢ Building homography matrix (created by the author)

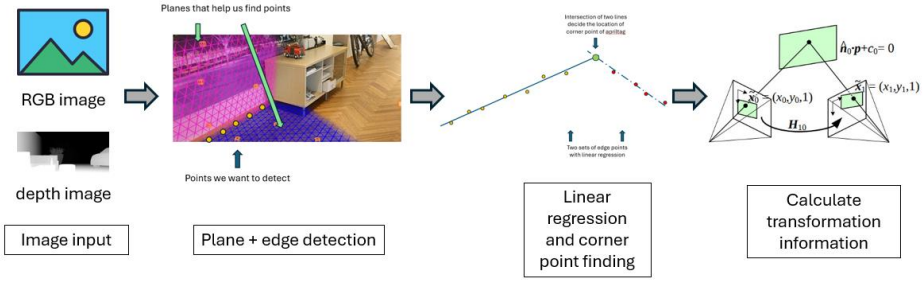


➢ Solving matrix elements (created by the author)

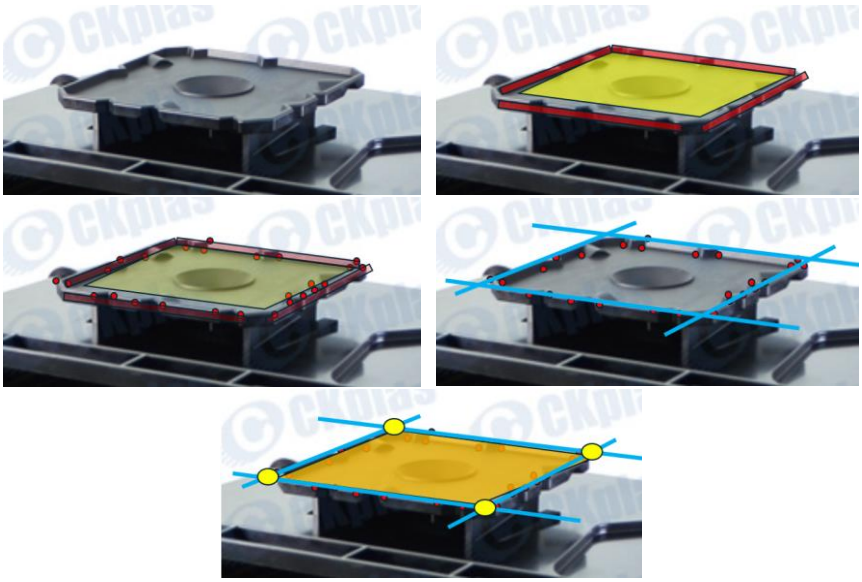
According to the data provided in [2], AprilTag guarantees good precision compared to other methods in repetitive and well-illuminated environments, such as factories. This is mainly due to its analytical approach to solving the problem. Therefore, I plan to continue using homography for its outstanding accuracy, while combining computer vision to find the corners of virtual AprilTags.

4 Proposed method

To replace AprilTag, I propose a method that combines the high precision of homography with the convenience of computer vision. First, to improve edge detection, we acquire an RGB-D image from a depth camera, using both color and depth information. Second, we locate edge points by marking plane junctions detected in the RGB-D image. Third, we perform linear regression to find the edges and corners of the FOUP flanges, which act as a virtual AprilTag. This step avoids potential errors from directly choosing corners and could be replaced by neural networks in the future once a sufficient dataset is collected. Finally, we compute the homography matrix elements to determine the analytical solution, rather than guessing poses.



➢ Task workflow (created by the author)



➢ Demonstrate of workflow (created by the author)

5 References

[1] Z. He, W. Feng, X. Zhao, and Y. Lv, "6D Pose Estimation of Objects: Recent Technologies and Challenges," *Applied Sciences*, vol. 11, no. 1, pp. 228, 2021. https://doi.org/10.3390/app11010228

[2] E. Olson, "AprilTag: A robust and flexible visual fiducial system," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2011, pp. 3400-3407.